

TP Machine Learning – Lab Report

Part 1 :

Test Prediction Error: 0.05

Sklearn Prediction Error (with K=1): 0.0

Test with K=1:

Prediction error for K=1: 0.04

The error decreased to 4% when using K=1, showing a slight improvement.

Test with K=5:

Prediction error for K=5: 0.0333

The error reduced to about 3.33%, indicating that using more neighbors can improve the model's accuracy.

Which means that Increasing K improves prediction accuracy by considering more neighbors.

Part 2 :

The results from the different classification algorithms show strong performance across the board. The SVM and Decision Tree models both achieved perfect accuracy with a prediction error of 0.0, indicating flawless classification of the test data.

The Naive Bayes model, initially showing a slight error of 0.022, performed well with only one misclassification. After fine-tuning the Naive Bayes model by adjusting the priors, the prediction error was reduced to 0.0, matching the performance of the other models. The confusion matrices for all models demonstrate that the predictions were almost perfect, with very few misclassifications. The best parameters found for the Naive Bayes model were {'priors': [0.4, 0.4, 0.2]}, showing how tuning can improve model performance.